

[BUG] First-order derivative with unshifted shift term in gradient recipe bre

<https://github.com/PennyLaneAI/pennylane/issues/2833>

ROW 95

[BUG] First-order derivative with unshifted shift term in gradient recipe breaks param_shift #2833

[New issue](#)[Closed](#)[#2834](#)

dwierichs opened on Jul 19, 2022

...

Expected behavior

The gradient is computed correctly by `qml.gradient.param_shift` even when using an operation that contains an unshifted term in its first-order gradient rule, provided via `gradient_recipe`.

Actual behavior

A custom operation with the gradient recipe containing the term `[coeff, 1, 0]` breaks `param_shift` (`qnode1` in the code below) unless all operations require such a term (`qnode2` in the code below)

Additional information

This is due to wrong handling of `unshifted_coeffs` in `expval_param_shift`. Will fix.

Source code

```
dev = qml.device('default.mixed', wires=2)

class OP(qml.operation.Operation):
    num_wires = 1
    ndim_params = (0,)
    num_params = 1
    grad_method = "A"
    #the first term corresponds to the unshifted tape
    grad_recipe = ([[1, 1, 0], [-1, 0, 1]],)

    @staticmethod
    def compute_decomposition(x, wires):
        return [qml.RX(x, wires=wires)]

@qml.qnode(dev)
def qnode1(params):
    OP(params[0], wires=0)
    qml.RY(0.1, wires=0)
    qml.RZ(params[1], wires=0)
    return qml.expval(qml.PauliZ(0))

@qml.qnode(dev)
def qnode2(params):
    OP(params[0], wires=0)
    OP(params[1], wires=0)
    return qml.expval(qml.PauliZ(0))

params = np.array([0.2, 0.3])
qnode(params)
qml.gradients.param_shift(qnode)(params)
```



Tracebacks

```
-----
IndexError                                Traceback (most recent call last)
Input In [6], in <cell line: 27>()
    25 params = np.array([0.2, 0.3])
    26 qnode(params)
--> 27 qml.gradients.param_shift(qnode)(params)

File ~/venvs/xanadu3.10/lib/python3.10/site-packages/PennyLane-0.25.0.dev0-py3.10.egg/pennylane/gradients/grac
    262     warnings.warn(
    263         "Attempted to compute the gradient of a QNode with no trainable parameters. "
    264         "If this is unintended, please add trainable parameters in accordance with "
    265         "the chosen auto differentiation framework."
    266     )
    267     return ()
--> 269 qjac = _wrapper(*args, **kwargs)
    271 if not hybrid:
    272     return qjac
```



Assignees

No one assigned

Labels

[bug](#)

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

[↗](#) Restructure 'param_shift' and fix bug in unshifted term bookkeeping
PennyLaneAI/pennylane

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Participants



```

272     return qjac

File ~/venvs/xanadu3.10/lib/python3.10/site-packages/PennyLane-0.25.0.dev0-py3.10.egg/pennylane/transforms/bat
300     gradient_fn = None
302     res = qml.execute(
303         tapes,
304         device=qnode.device,
305     )
306     (**execute_kwargs,
307     )
--> 313     return processing_fn(res)

File ~/venvs/xanadu3.10/lib/python3.10/site-packages/PennyLane-0.25.0.dev0-py3.10.egg/pennylane/gradients/para
267     g = qml.math.tensordot(res, qml.math.convert_like(gradient_coeffs[i], res), [[0], [0]])
269     if unshifted_coeffs:
270         # add on the unshifted term
--> 271         g = g + unshifted_coeffs[i] * r0
273     grads.append(g)
275 # The following is for backwards compatibility; currently,
276 # the device stacks multiple measurement arrays, even if not the same
277 # size, resulting in a ragged array.
278 # In the future, we might want to change this so that only tuples
279 # of arrays are returned.

IndexError: list index out of range

```

System information

Name: PennyLane
Version: 0.25.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: <https://github.com/XanaduAI/pennylane>
Author:
Author-email:
License: Apache License 2.0
Location: /home/david/venvs/xanadu3.10/lib/python3.10/site-packages/PennyLane-0.25.0.dev0-py3.10.egg
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, retworkx, scipy, semar
Required-by: PennyLane-Lightning

Platform info: Linux-5.4.0-121-generic-x86_64-with-glibc2.31
Python version: 3.10.5
Numpy version: 1.23.1
Scipy version: 1.9.0rc2
Installed devices:
- lightning.qubit (PennyLane-Lightning-0.24.0)
- default.gaussian (PennyLane-0.25.0.dev0)
- default.mixed (PennyLane-0.25.0.dev0)
- default.qubit (PennyLane-0.25.0.dev0)
- default.qubit.autograd (PennyLane-0.25.0.dev0)
- default.qubit.jax (PennyLane-0.25.0.dev0)
- default.qubit.tf (PennyLane-0.25.0.dev0)
- default.qubit.torch (PennyLane-0.25.0.dev0)

Existing GitHub issues

☒ I have searched existing GitHub issues to make sure the issue does not already exist.



dwierichs added **bug** on Jul 19, 2022



dwierichs mentioned this on Jul 19, 2022

[Restructure param_shift and fix bug in unshifted term bookkeeping #2834](#)



dwierichs closed this as **completed** in [#2834](#) on Jul 21, 2022

Restructure param_shift and fix bug in unshifted term bookkeeping #2834

Merged dwierichs merged 13 commits into master from restructure-param_shift on Jul 21, 2022

Conversation 33 Commits 13 Checks 0 Files changed 3 +153 -90

dwierichs commented on Jul 19, 2022

This PR restructures param_shift . In particular, it introduces additional helper methods, trying to modularize the main for loop of expval_param_shift . In addition, this restructuring fixed #2833, as this bug was caused by flawed bookkeeping of unshifted coefficients (it worked for second order derivatives because usually then all operations require the unshifted tape.

Related issue: #2833

dwierichs added 5 commits 3 years ago

- restructure
- tests
- black
- changelog
- restructure

codecov bot commented on Jul 19, 2022 • edited

Codecov Report

Merging #2834 (9068c6f) into master (cfef693) will increase coverage by 0.00% . The diff coverage is 100.00% .

	Coverage	Diff	
@@	master	#2834	+/-
Coverage	99.62%	99.62%	
Files	250	250	
Lines	20517	20521	+4
+ Hits	20441	20445	+4
Misses	76	76	

Impacted Files	Coverage Δ
pennylane/gradients/parameter_shift.py	100.00% <100.00%> (0)

Continue to review full report at Codecov.

Legend - Click here to learn more
Δ = absolute <relative> (impact), 0 = not affected, ? = missing data
Powered by Codecov. Last update cfef693...9068c6f. Read the comment docs.

Reviewers

- josh146
- mlxd
- antalszava

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[BUG] First-order derivative with unshifted s...

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4 participants

- josh146
- mlxd
- antalszava
- dwierichs